

Car Rental Booking Service

Instructions for the Candidate

- All details provided in this assignment are sufficient. You may make additional logical assumptions when needed.
 - Deliver **production-ready Spring Boot code**.
 - The code must **build successfully** using Maven or Gradle.
 - Deliver the solution via **GitHub**.
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Assignment

Velocity Motors, a global mobility company, uses a microservice-based architecture for its digital ecosystem. One of their microservices, **car-booking-service**, is responsible for managing car rental bookings.

Your task is to build the following functionalities inside a **Spring Boot application**:

Functional Requirements

1. REST endpoint to confirm a car booking

Booking confirmation depends on the **payment method**:

- **If payment mode is *digital wallet***
→ Booking must be confirmed immediately.
- **If payment mode is *credit card***
→ Call REST API of **credit-card-validation-service** to verify payment status.
 - If validation = *APPROVED*, confirm the booking.
 - Else, throw an error.
- **If payment mode is *bank transfer***
→ Booking must be created with status **PENDING_PAYMENT**.
- The service must also consume the event topic **bank-transfer-payment-events** to listen for updates and change the booking status to **CONFIRMED** when full payment is received.

2. Automatic cancellation

If a booking was paid by **bank transfer**, and the **full amount is not received at least 48 hours before the rental start date**, the booking must be **automatically cancelled**.

REST Endpoint Details

1. Confirm Car Booking

Request Data

- Customer Name
- Vehicle ID
- Rental Start Date
- Rental End Date
- Vehicle Category
 - *Compact*
 - *Sedan*
 - *SUV*
 - *Luxury*
- Payment Mode
 - *Cash*
 - *Credit Card*
 - *Bank Transfer*
- Payment Reference

Response

- Booking ID
- Booking Status
 - **PENDING_PAYMENT**
 - **CONFIRMED**
 - **CANCELLED**

Validations

- A vehicle cannot be booked for more than **21 days**.
- Rental end date must be after the rental start date.
- Vehicle ID must be valid (mock or assume validation logic).

Additional Information

Event-Driven Architecture

Velocity Motors uses Kafka-based event-driven processing.

The topic **bank-transfer-payment-events** publishes payment completion events.

Event Structure

Attribute	Description
paymentId	Unique identifier of payment
senderAccountNumber	Account number of the sender
paymentAmount	Amount received
transactionDetails	Format: <TxnRef(12 chars)> <BookingId(10 chars)> Example: TXN987654321 BKG0012345

Credit Card Validation Service Specs

The OpenAPI spec file **credit-card-validation-service.yaml** is provided in the attachment.

Your service must integrate and use it.